

# Thursday Warm-up: Magnetic Vector Potential and Magnetic Dipoles

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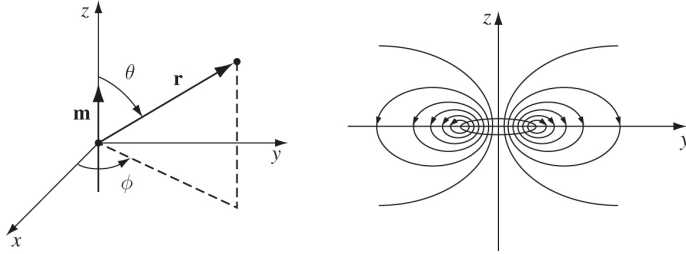


Figure 1: (Left) A magnetic dipole moment oriented along the z-axis. (Right) The  $\mathbf{B}$ -field of a magnetic dipole.

Express the vector potential as a function of  $r$  and  $\theta$ .

## 1 Memory Bank

- **Magnetic vector potential:**

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (1)$$

- **Magnetic flux and vector potential:**

$$\Phi_B = \int \mathbf{B} \cdot d\mathbf{a} = \oint \mathbf{A} \cdot d\mathbf{l} \quad (2)$$

- Curl in spherical coordinates,  $\mathbf{A}(r, \theta) = A(r, \theta)\hat{\phi}$

$$\nabla \times \mathbf{A} = \frac{1}{r \sin \theta} \left( \frac{\partial}{\partial \theta} (\sin \theta A_\phi) \right) \hat{r} - \frac{1}{r} \left( \frac{\partial}{\partial r} (r A_\phi) \right) \hat{\theta} \quad (3)$$

- (a) Calculate the  $\mathbf{B}$ -field of the dipole in Fig. 1, using the curl in spherical coordinates. (b) Sketch the dipole field, and show that it has the structure of an electric dipole field (Fig. 1).

- Suppose a square loop of current with side length  $d$  lies in the xy-plane centered at the origin, in the presence of a uniform  $\mathbf{B}$ -field in the  $\hat{x}$  direction. (a) What is the magnetic dipole moment,  $\mathbf{m}$ ? (b) Show that the torque on the loop is

$$\mathbf{N} = \mathbf{m} \times \mathbf{B} \quad (6)$$

## 2 Magnetic Vector Potential and Magnetic Dipoles

- The magnetic dipole moment of a loop of current is define like:

$$\mathbf{m} = I \int d\mathbf{a} \quad (4)$$

- Suppose a circular loop of current with radius  $b$  lies in the xy-plane, centered at the origin (Fig. 1). Calculate  $\mathbf{m}$ .
- According to the multipole expansion of the magnetic vector potential, the dipole component of the vector potential is

$$\mathbf{A}_{\text{dip}} = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \hat{\mathbf{r}}}{r^2} \quad (5)$$